

John Liu

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EDUCATION & AWARDS

University of Waterloo

Computer Science, 3.9/4.0 Major GPA

Waterloo, ON

Expected April 2028

- Canadian Mathematical Olympiad (CMO) Qualifier, AIME Qualifier (5×)

WORK EXPERIENCE

Figma

Software Engineer Intern

San Francisco, CA

May 2026 – Aug 2026

- Implemented a **C++/WebAssembly** evaluation engine for resolving variable expressions across dependency graphs containing **60K+** relationships, including cycle detection and performant dependency resolution
- Shipped **design token compositions** and expressions to enterprise customers, introducing a programmable variable primitive for large-scale design systems
- Reduced memory usage by **18%** for composed design variables by avoiding duplicated paint storage across canvas nodes and lazily resolving aliases at rendering time

WindBorne Systems Inc.

Software Engineer Intern

Palo Alto, CA

May 2025 – Aug 2025

- **Employee #5** on the product engineering team, working across forecasting, data infrastructure, and visualization systems for energy trading products at a Khosla-backed weather technology startup
- Owned development for a meteorological insights platform used by energy traders, building features across **TypeScript, Rails, Rust, and WebGL** systems supporting a product generating **\$2.5M ARR**
- Implemented tropical cyclone forecasting algorithms in **C++** and **Python** over **200TB+** of monthly global weather data, reducing query latency by **28%** through a custom geospatial caching layer
- Built performant weather visualization features for a **Rust/WebGL** graphing engine, maintaining interactive frame rates while rendering millions of forecast data points in the browser

CIBC

Software Engineer Intern

Toronto, ON

Jan 2025 – Apr 2025

- Delivered real-time chatbot features using **Python, Redis**, and websocket streaming, enabling low-latency tool execution and interactive user experiences for CIBC's internal AI assistant
- Engineered LLM-powered capabilities including **RAG** over patent databases, live news ingestion pipelines, and sentiment classification systems for document triage and automated summarization
- Architected a full-stack LLM prototyping framework (**Next.js, Prisma, FastAPI**) adopted across **10+** internal repositories, standardizing deployment and evaluation workflows

PROJECTS

unfollowed.lol | JavaScript, PostgreSQL, Django

- Launched a Chrome extension for tracking marketing analytics that reached **#8** on Product Hunt with **100+** upvotes and reached **4,000+** daily active users

SlimeScholars | React, Next.js, MongoDB, AWS

- Built an educational content platform with a custom lesson builder supporting **LaTeX, XML**, and custom markdown-based authoring workflows, with analytics dashboards tracking student engagement across lessons

Marble Investments | TypeScript, React, Next.js, MongoDB

- Created a **Stripe**-powered investment platform supporting fund subscriptions, redemptions, and investor onboarding with elliptic-curve cryptography for sensitive user data

SKILLS

Languages: Python, C++, TypeScript, Rust, Go, Java, Ruby, SQL

Frameworks: React, Next.js, Rails, FastAPI, PostgreSQL, Docker, Kubernetes, Redis, MongoDB, AWS

Areas: Operating Systems, Distributed Systems, WebGL, Algorithms & Data Structures, Networking, Concurrency